O–I–501b
January 2, 1964

SUPERSEDING
Fed. Spec. O-I-501a
June 25, 1953

FEDERAL SPECIFICATION

INHIBITORS, PICKLING (FOR USE WITH SULPHURIC ACID)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope This specification covers inhibitors which when added to sulfuric acid pickling baths diminish the attack of acid on descaled areas without appreciably retarding the descaling rate.

1.2 Classification.

1.2.1 *Types and classes.* Pickling inhibitors shall be of the following types and classes, as specified (see 6.1):

Types:

I—Foaming.
II—Nonfoaming.

Classes:

a—Liquid.
b—Solid.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-S-22698 — Steel Plate, Carbon, Structural, for Ships.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

FSC 6850

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(Copies of Military specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations:

49 CFR 71-78—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(Copies of the Code of Federal Regulations may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders should cite "49 CFR 71-78 (Rev. 1956).")

3. REQUIREMENTS

3.1 **Descaling time.** When tested as specified in 4.3.2, the ratio $\frac{T_1}{T_0}$ shall not exceed 1.35 and the ratio $\frac{T_2}{T_0}$ shall not exceed 1.8.

3.2 **Hydrogen evolution.** When tested as specified in 4.3.3, the ratios $\frac{V_1}{V_0}$ and $\frac{V_2}{V_0}$ shall not exceed 0.15.

3.3 **Staining and filming of steel.** The inhibited acid solution shall not stain the steel, and shall not leave films which would interfere with galvanizing, tinning, painting, electroplating, or other processes used for coating steel, in any of the tests specified in 4.3.2.2 and 4.3.2.3.

3.4 **Arsenic (as As).** Inhibitors shall not contain more than 0.10 percent of arsenic when tested as specified in 4.3.4.

3.5 **Higher strength inhibitors.** When products, two, three, or more integral times the specified strength are offered, a corresponding reduction in the quantities of inhibitor shall be made when subjecting to the tests specified in section 4.

3.5.1 *Dilution ratio.* This ratio shall be equal to the reciprocal of the strength value.

3.6 **Type I, foaming.** Type I shall be compounded to produce a complete blanket of foam over the surface of pickling baths when tested as specified in 4.3.2 and 4.3.3. The foam blanket shall be effective in reducing acid fumes when used in pickling baths under the conditions of bath concentration and temperature specified in 4.3.2 and 4.3.2.2, but shall not foam to such an extent that the pickling baths might foam over when the acid solution is within 3 inches of the top of the container.

3.7 **Type II, nonfoaming.** Type II shall not produce foam when tested as specified in 4.3.2 and 4.3.3.

3.8 **Class a, liquid.** Class a shall not be so viscous as to cause difficulty in handling and shall not contain volatile solvents nor evolve flammable or obnoxious vapors either alone or on mixing with the pickling bath.

3.9 **Class b, solid.** Class b shall not form a hard cake in the container or pickling vat, nor produce irritation or poisonous dust before solution nor evolve poisonous or obnoxious vapors after solution, and shall be in the form of a free-flowing powder.

3.10 **Workmanship.** The product shall be manufactured of best quality materials, and in such manner as to produce an inhibitor capable of uniformly dissolving in sulfuric acid pickling baths to yield maximum effectiveness immediately on mixing.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection.

4.2.1 *Lot.* For purposes of sampling, a lot shall consist of all material manufactured as one batch.

4.2.2 *Tests.* From each lot, two containers shall be selected at random. From each of the two containers, 1-pound specimens shall be taken and placed in separate clean, dry, metal or glass containers, sealed, marked, and forwarded to a laboratory satisfactory to the procuring activity for the tests specified in 4.3.

4.2.3 *Examination of filled containers.* A random sample of filled containers shall be taken in accordance with MIL-STD-105 at inspection level I and acceptable quality level = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be cause for rejection of the nonconforming container, and if the number of nonconforming containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, this shall be cause for rejection of the lot represented by the sample.

4.3 Test procedures.

4.3.1 *Specimens of steel.* The inhibitor shall be tested with 1 by 5-inch specimens cut from 1/8-inch hot-rolled steel sheet, conforming to type I, class A of MIL-S-22698. These specimens shall bear a uniform mill scale, and shall be cut from at least two different sheets which shall be suitably identified.

4.3.2 *Descaling time.*

4.3.2.1 *With uninhibited acid solution* (T_0). The uninhibited acid solution shall contain 50 ml. of 66 degrees Baume sulfuric acid per liter. Approximately 150 ml. of the acid solution shall be placed in a 1.5 by 8-inch pyrex test-tube and maintained at 180° ± 1°F. in a thermostatically controlled bath. A steel specimen shall be immersed in the acid solution in the tube for 4 minutes. The specimen shall then be removed, rinsed immediately with tap water, then alcohol, and then thoroughly dried, and the degree of scale removal evaluated with a low-power microscope. The test shall be repeated, using a fresh steel specimen and fresh acid solution and increasing or decreasing the exposure time until t_0, the minimum time required to descale the steel completely, is determined to the nearest 0.5 minute. The test shall be repeated using fresh acid solution and steel specimens from a different sheet. The average of the 2 values of t_0 found shall be taken as the descaling time of the uninhibited acid, T_0.

4.3.2.2 *With unaged, inhibited acid* (T_1). The test specified in 4.3.2.1 shall be repeated, using a fresh acid bath containing (0.6 x dilution ratio) grams of the inhibitor to be tested per liter of sulfuric acid solution specified in 4.3.2.1. A weighing bottle shall be used in weighing the required quantity of inhibitor. Let T_1 equal the average of the individual descaling times. The specimens shall be examined for staining and filming.

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4.3.2.3 *With aged, inhibited acid* (T_2). A test solution shall be prepared by mixing 800 ml. of water, 300 g. of ferrous sulfate ($FeSO_4 .7H_20$), 50 ml. of 66 degrees Baume sulfuric acid (0.6 x dilution ratio) grams of inhibitor and diluting the resulting mixture to one liter. One liter of the test solution shall be placed in a 2-liter flask under a water cooled reflux condenser and boiled for 3 hours. Approximately 150 ml. of the test solution shall be placed in a 1.5-inch by 8-inch pyrex test tube and maintained at 180° ± 1°F. in a thermostatically controlled bath. A steel specimen shall be immersed in a bath of the solution for 4 minutes, then removed, rinsed immediately with tap water, then alcohol, and then thoroughly dried, and the degree of scale removal evaluated with a low power microscope. This test shall be repeated, using another bath of the solution and a fresh steel specimen and increasing or decreasing the exposure time until t_2, the minimum time required to descale the steel completely is determined to the nearest 0.5 minute. The test shall be repeated using new specimens taken from a different sheet and fresh test solution. The average of the 2 values of t_2 found shall be taken as the descaling time of the aged, inhibited acid, T_2. The specimens shall be examined for staining and filming.

4.3.3 *Hydrogen evolution.*

4.3.3.1 *Procedure with uninhibited acid* (V_0). One-hundred-fifty ml. of uninhibited acid solution (see 4.3.2.1) shall be placed in a 1.5-inch by 8-inch pyrex test tube and maintained at a temperature of 130° ± 1.0° F., with a thermostatically controlled bath. A smooth, clean steel specimen previously descaled in uninhibited acid solution shall be immersed in the solution and the test tubes shall be sealed immediately with a stopper bearing a discharge tube. The evolved hydrogen shall be collected at room temperature by water displacement from a graduated cylinder of suitable size and all volumes shall be converted to the corresponding volume for standard conditions of temperature and pressure (0°C. and 760 mm., respectively). The gas volume shall be read at 10-minute intervals to detect significant changes in rate of solution and the total volume (ml.) collected during 30 minutes shall be taken as the gas volume (v_0) of the steel being used in the test. This test shall be repeated, using a fresh bath and a steel specimen from a different source. The average of v_0 volumes shall be taken as the hydrogen evolution value of the uninhibited acid, V_0. Triplicate determinations shall be made for each steel specimen, using a fresh bath each time.

4.3.3.2 *Procedure with unaged inhibited acid* (V_1). The tests specified in 4.3.3.1 shall be repeated, using test steels from the same sheets and inhibited acid prepared as specified in 4.3.2.2. The average of v_1 volumes shall be taken as the hydrogen evolution value of unaged inhibited acid, V_1.

4.3.3.3 *Procedure with aged inhibited acid* (V_2). The tests specified in 4.3.3.1 shall be repeated, using test steel specimens from the same sheet and aged inhibited acid prepared as specified in 4.3.2.3. The average of v_2 volumes shall be taken as the hydrogen evolution value of aged inhibited acid, V_2.

4.3.4 *Arsenic content.* Place 10.0 g. of the sample in a 200-ml. flask. Add 5 g. cuprous chloride and 75 ml. HCl (sp. gr. 1.19). Mix well, insert a thermometer and arrange the flask and a condenser for downward distillation.[1] Distill approximately 35 ml. of the solution into 150 ml. of cold water contained in a 400-ml. beaker cooled in an ice or cold-water bath. (The condenser tip should dip below the surface of the water in the beaker. The distillation should be watched carefully to avoid suck-back.) The temperature of the vapor during distillation must remain below 108°C. otherwise halt the distillation, cool the flask, and add 35 ml.

[1] The rubber stoppers used should be freed of sulfur by washing well with carbon disulfide followed by boiling with 1.1 HCl for a few minutes. Allow to dry before using.

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of concentrated HCl before continuing the distillation. Neutralize the distillate carefully with 25 percent NaOH solution, then add 1:1 HCl until just acid. Add 15 to 20 ml. of cold saturated $NaHCO_3$ solution.

Add 1 g. of KI crystals and 5 ml. of 1 percent starch solution. Stir until the KI is dissolved and titrate with 0.01*N* iodine solution.

$$\text{Percent arsenic} = \frac{3.75\ AN}{W}$$

where:

A = ml. of iodine solution used.
N = normality of iodine solution.
W = weight of sample.

5. PREPARATION FOR DELIVERY

(For civil agency procurements Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Packing.

5.1.1 *Levels A and B.*

5.1.1.1 *For class a, liquid.* Class a inhibitors shall be furnished in 1-gallon or 5-gallon glass carboys, conforming to Specification 1A of the Code of Federal Regulations.

5.1.1.2 *For class b, solid.* Class b inhibitors shall be furnished in 50-, 100-, or 200-pound fiberboard drums, conforming to type I of PPP-D-723.

5.1.2 *Level C.* Inhibitors shall be packed in containers in a manner which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Military agencies.* In addition to the marking specified in 5.2.3 and any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

5.2.2 *Civil agencies.* In addition to any special marking required by the contract or order and 5.2.3 containers shall be marked in accordance with Fed. Std. No. 123.

5.2.3 *Labeling.* The exterior containers shall be durably and legibly labeled with complete directions on the use of inhibitor in plant pickling practice. The dilution ratio shall be displayed on the label in prominent numerals. Directions shall include a table showing the amount of inhibitor required per 100 gallons of pickling solution for sulfuric acid concentrations normally used.

6. NOTES

6.1 Intended use. The inhibitors covered by this specification are to be used in the pickling of steel plate forgings and other steel articles. These inhibitors are long-life, heat-stable products, and care should be taken to avoid excessive inhibitor build-up in the pickling bath and consequent long pickling cycles. They should be used only in conjunction with sulfuric acid, and should be used in the dilution as recommended by the manufacturers.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of inhibitor required (see 1.2).
(*c*) Size of container required (see 5.1).

6.3 Invitations for bids should specify that bidders shall include in their bids the dilution ratio in which the product will satisfy the requirements of this specification. The bid price will be multiplied by this ratio and the results compared for determination of the low bid in making the award.

6.4 Steel should be aged at least 24 hours after pickling before fabrication or welding to prevent any deleterious effects due to hydrogen embrittlement.

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6.5 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Chemicals, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Chemicals, not otherwise indexed.
Truckload minimum weight 24,000 pounds subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—MU

Navy—Sh

Air Force—AFSC (ASD)

PREPARING ACTIVITY:

Navy—Sh

Copies of this specification may be purchased for 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863. WHICH IS OBSOLETE PLATE NO. 15419

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